University of Nottingham
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COMP3089 Data Visualization Project

Project Title : *Interactive Data Visualisation of Global and Regional Sea Level Rise (1993–2024)*

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1. Introduction

Climate change has become one of the most critical global challenges in the modern world. Rising global temperatures have caused significant environmental changes, including melting glaciers, increasing ocean temperatures, and rising global sea levels. Among these impacts, sea level rise is considered one of the most important indicators of climate change because it directly reflects changes in the Earth's climate system.

Global sea level rise occurs primarily due to two main factors: thermal expansion of seawater as ocean temperatures increase and the melting of glaciers and polar ice sheets. As the planet warms, ocean water expands and contributes to higher sea levels. Additionally, melting ice from Greenland and Antarctica adds more water to the oceans, further increasing sea levels.

Rising sea levels pose significant risks to coastal communities, infrastructure, and ecosystems. Even small increases in ocean levels can lead to coastal flooding, shoreline erosion, and saltwater intrusion into freshwater resources. Therefore, monitoring and analysing sea level changes is important for understanding climate change and predicting its future impacts.

The goal of this project is to develop an interactive data visualisation dashboard that allows users to explore global and regional sea level trends between 1993 to 2024. The dashboard provides multiple visualisations that enable users to analyse sea level trends, yearly variations, and regional differences across major ocean basins.

The objectives of this project include:

- Visualising long-term global sea level trends
- Analysing yearly changes and variability in sea level measurements
- Comparing sea level trends across major ocean regions
- Providing interactive tools that allow users to explore climate data

The target audience of this project includes students, researchers and the general public who are interested in understanding climate change trends throughout visual data analysis. By transforming raw datasets into interactive visualisations, this project aims to improve accessibility and understanding of climate-related information.

2. Related Work

Numerous scientific studies have investigated the causes and consequences of global sea level rise. According to Mimura (2013), sea level rise is one of the most direct consequences of climate change and is closely linked to rising global temperatures and melting polar ice caps.

Satellite observations have significantly improved the accuracy of sea level measurements. Since the early 1990s, satellite altimetry has enabled scientists to monitor global seal level with high precision. Research indicates that global sea levels have been rising steadily for the past three decades and that the rate of increase has accelerated in recent years. Hamilton et al. (2024) reported that global mean sea level rise has accelerated significantly over the past three decades, increasing from approximately 2mm per year in early 1990s to over 4mm per year in recent years. This acceleration highlights the increasing impact of climate change on ocean systems.

In addition to global trends, regional variations in sea level rise have also been observed very closely. Ocean circulation patterns, wind systems, and regional climate conditions can cause uneven distribution of sea level changes across different ocean basins. For example, some regions experience higher rates of sea level rise due to ocean current and atmospheric interactions.

Data visualisation has become an important tool for analysing and communicating climate data. Interactive dashboards allow users to explore complex datasets and identify patterns that may not be immediately visible through raw data analysis. Tools such as Python, Plotly, and Streamlit enable developers to build dynamic visualisations that allow users to interact with climate datasets in real time.

This project builds upon these concepts by combining multiple visualisation techniques into a single interactive dashboard that allows users to explore both global and regional sea level trends.

3. Methods and Design

3.1. Dataset Description

The datasets used in this project were obtained from publicly available sources including NASA, NOAA, and Kaggle. The primary dataset consists of global mean sea level measurements from 1993 to 2024, derived from satellite altimetry. Satellite data provides high temporal consistency and global coverage, making it suitable for analysing long-term climate trends.

The dataset includes key variables such as:

- Year
- Sea level measurements (mm)
- Smoothed sea level values
- Yearly rate of change
- Cumulative sea level rise

In addition, regional datasets for the Atlantic, Pacific, and Indian Oceans were used to examine spatial variability. These derived features (e.g., moving averages, cumulative rise) were calculated to enhance interpretability and reduce noise in the raw measurements.

3.2. Visualisation Design

Five different visualisation techniques were used in the dashboard to present different perspectives of the dataset

3.2.1. Line Chart - Global Trend

A line chart is the most appropriate choice for time-series data as it preserves temporal continuity and allows users to clearly observe long-term trends. Compared to bar charts, line charts avoid visual fragmentation and make it easier to identify gradual increases in sea level over time.

3.2.2. Bar Chart - Yearly Variability

The bar chart is used to represent year-to-year changes in sea level. This visualisation is effective for highlighting short-term fluctuations and distinguishing between positive and negative changes. Unlike a line chart, the bar chart emphasises magnitude differences relative to zero, making variability more interpretable.

3.2.3. Histogram - Distribution Analysis

The histogram illustrates the distribution of sea level values across the dataset. This allows users to observe how measurements are concentrated and whether the distribution shifts over time. It is particularly useful for identifying whether higher sea level values are becoming more frequent, which cannot be easily seen in time-series charts.

3.2.4 Heat Map - Seasonal and Temporal Patterns

The heatmap visualises sea level data across both months and years. This dual-axis representation enables the identification of seasonal variations while simultaneously revealing long-term trends. Compared to multiple line charts, the heatmap provides a more compact and intuitive overview of complex temporal patterns.

3.2.5 Area Chart - Cumulative Impact

The area chart represents cumulative sea level rise over time. This visualisation emphasises how small yearly increases accumulate into significant long-term changes. It is particularly effective for communicating the long-term impact of climate change, as it highlights the continuous and irreversible nature of sea level rise.

These visualisation techniques were selected because they complement each other and provide multiple perspectives of the dataset.

4. Implementation

This project dashboard was implemented and built using Python and several data visualisation libraries. The main technologies used include Streamlit, Plotly, Pandas and NumPy.

- Pandas - used for data cleaning, transformation, and feature engineering
- NumPy - used for numerical computations such as trend estimation and moving averages
- Plotly - used to create interactive visualisations
- Streamlit - used to build the user interface and enable interactivity

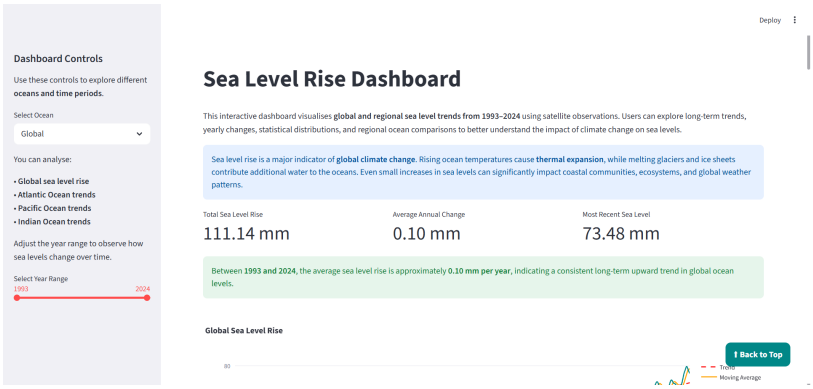


Fig 1
(Main Interface of the dashboard)

4.1 Data Preprocessing

The dataset was first cleaned by removing missing values and ensuring consistency in time formatting. Derived features such as yearly change, cumulative sea level rise, and moving averages were calculated to enhance analysis. This preprocessing step ensures that the data is structured appropriately for both time-series and statistical visualisations.

4.2. Visualisation Implementation

Multiple visualisations were developed using Plotly, with each chart designed to reveal a different aspect of sea level change while collectively providing a comprehensive understanding of the dataset.

The line chart serves as the primary visualisation, presenting the global sea level trend over time in a continuous and intuitive manner. This allows users to clearly observe long-term patterns and identify the overall upward trajectory. To enhance interpretability, trend lines and smoothed curves were incorporated, reducing short-term noise and making underlying patterns more apparent.

To complement this, a bar chart was used to represent yearly changes in sea level. Unlike the line chart, which emphasises continuity, the bar chart highlights variability between individual years, making it easier to compare the magnitude of increases and identify periods of higher or lower change.

A heatmap was introduced to provide a two-dimensional perspective of the data, displaying sea level values across both months and years. This enables the simultaneous analysis of seasonal variations and long-term trends, offering insights that would be less visible in single-axis visualisations.

In addition, a histogram was used to examine the statistical distribution of sea level values. This helps reveal how frequently different value ranges occur and whether the distribution shifts towards higher values over time, providing a deeper understanding beyond simple trend analysis.

Finally, an area chart was implemented to illustrate cumulative sea level rise. By emphasising the accumulation of yearly increases, this visualisation highlights the long-term impact of gradual changes and reinforces the progressive nature of sea level rise.

```
fig1 = px.line(
    filtered,
    x="YearPlusFraction",
    y="SmoothedGMSLWithGIA",
    title="Global Sea Level Rise",
    labels={
        "YearPlusFraction": "Year",
        "SmoothedGMSLWithGIA": "Sea Level (mm)"
    },
    color_discrete_sequence=["darkcyan"],
    hover_data={
        "Year":True,
        "Month":True,
        "SmoothedGMSLWithGIA":":.2f"
    }
)
```

Fig 2

An example of the implementation used to create the global sea level line chart.

This code creates a line chart that displays sea level measurements over time. Additional features such as trend lines and moving averages were added to highlight long-term patterns in the data.

4.3. Interactive Dashboard

Streamlit was used to integrate all visualisations into a unified interactive dashboard, allowing the different analytical components to function cohesively within a single interface. The dashboard provides user controls for selecting specific ocean regions, including the global view as well as the Atlantic, Pacific, and Indian Oceans, in addition to adjusting the time range for analysis.

These interactive features enable users to actively engage with the dataset rather than passively viewing static outputs. By dynamically filtering regions and time periods, users can more easily identify patterns, compare regional trends, and explore variations in sea level behaviour across different contexts. This flexibility enhances the interpretability of the visualisations, as insights can be tailored to specific perspectives of interest.

Overall, the integration of interactivity significantly improves both usability and analytical depth, transforming the dashboard from a static presentation tool into an exploratory platform that supports more meaningful data-driven insights.

5. Results and Discussion

The visualisations developed in this project provide a comprehensive overview of global and regional sea level trends between 1993 and 2024. By combining multiple visualisation techniques, the dashboard enables both macro-level and detailed analysis of sea level changes over time.

5.1. Global Sea Level Trends

The line chart (Figure 3) illustrates the long-term trend of global sea level rise from 1993 to 2024. A clear and consistent upward trajectory is observed throughout the entire time period, indicating that global sea levels have been steadily increasing over the past three decades. This continuous rise reflects the cumulative impact of climate-related processes such as thermal expansion of seawater and the melting of glaciers and ice sheets.

The inclusion of a regression trend line further reinforces this observation by highlighting the overall direction of change in a simplified linear form. The positive slope of the trend line confirms a sustained upward trend in sea levels over time. While the linear model does not explicitly capture acceleration, it provides a clear indication of the long-term increase and supports the interpretation that sea level rise is persistent and ongoing.

In addition to the regression line, a 12-month moving average is used to smooth short-term fluctuations in the data. This reduces noise caused by seasonal variability and measurement inconsistencies, allowing the underlying trend to be observed more clearly. While individual data points may show minor variations, the moving average reveals a stable and continuous upward pattern. This is particularly important in climate data analysis, where short-term variability can obscure long-term trends.

The combination of raw data, trend line, and moving average enhances the interpretability of the visualisation by providing multiple perspectives on the data. Together, these elements demonstrate that

sea level rise is not a temporary fluctuation but a sustained and long-term process. The consistency of this upward trend highlights the severity of climate change impacts and emphasises the importance of continued monitoring and mitigation efforts.

5.2. Yearly Variability in Sea Level Change

The bar chart *(Figure 4)* presents the yearly changes in sea level measurements, offering insight into short-term variability within the overall trend. Unlike the line chart, which focuses on cumulative patterns over time, this visualisation highlights how sea levels fluctuate from one year to the next.

The data shows that annual sea level changes are not uniform. Some years exhibit larger increases, while others show smaller increments or slight decreases. These variations can be attributed to natural climate phenomena such as El Niño and La Niña, which influence ocean temperature, atmospheric circulation, and precipitation patterns.

Despite these short-term fluctuations, a key observation is that the majority of yearly changes are positive. This indicates that increases in sea level consistently outweigh any temporary decreases. Even in years where the increase is smaller, the overall contribution to long-term sea level rise remains positive.

This pattern highlights an important aspect of climate data: variability does not contradict long-term trends. Instead, it reflects the complexity of the Earth's climate system, where short-term environmental factors interact with long-term global processes. Therefore, while the yearly changes may appear inconsistent, they collectively contribute to a steady upward trend in sea levels.

5.3. Distribution of Sea Level Values

The histogram *(Figure 5)* provides a statistical representation of the distribution of sea level values across the dataset. The distribution is skewed towards higher values, indicating that more recent measurements tend to have greater sea level heights compared to earlier years.

The mean sea level, indicated by the dashed vertical line, is positioned closer to the upper range of the dataset. This suggests that a significant portion of the data consists of higher sea level values, which reflects the progressive increase observed over time.

Lower sea level values are less frequent and are primarily associated with earlier years in the dataset. As time progresses, the dataset becomes increasingly dominated by higher values, which shifts the overall distribution to the right. This shift is a strong indicator of long-term change and supports the conclusion that sea level rise is cumulative.

Furthermore, the shape of the histogram demonstrates how the distribution evolves over time. Instead of being evenly spread, the concentration of values in higher ranges indicates that sea level rise is not only occurring but becoming more pronounced in recent years.

This visualisation reinforces the understanding that sea level rise is a gradual but continuous process, with increasing frequency of higher values over time.

5.4. Seasonal and temporal patterns

The heatmap *(Figure 6)* provides a two-dimensional representation of sea level values across both months and years. This allows for the analysis of seasonal variations as well as long-term temporal trends within the dataset.

One of the most noticeable features of the heatmap is the gradual increase in colour intensity over time. As the years progress, the colours become darker, indicating higher sea level values in more recent periods. This visual pattern clearly reflects the long-term upward trend observed in the line chart.

Seasonal variations can also be observed across different months. However, these variations are relatively small compared to the overall increase in sea levels. While certain months may show slightly higher or lower values due to environmental factors, the dominant pattern remains the consistent rise in sea levels over time.

This suggests that although seasonal effects exist, they do not significantly alter the long-term trend. Instead, they are secondary to the broader impact of climate change.

The heatmap is particularly effective in visualising how sea level changes occur across both time scales simultaneously. It allows users to identify patterns that may not be immediately visible in single-dimensional charts, making it a valuable tool for exploring complex datasets.

5.5. Cumulative sea level rise

The area chart *(Figure 7)* illustrates the cumulative increase in sea level since the beginning of the dataset. This visualisation emphasises how small yearly changes accumulate over time to produce significant long-term effects.

The curve shows a steady upward progression, indicating continuous accumulation of sea level rise. Unlike the yearly change chart, which focuses on short-term variations, this chart highlights the long-term impact of these changes.

The slope of the curve appears to increase slightly in more recent years, suggesting a potential acceleration in sea level rise. This observation aligns with scientific research that indicates the rate of sea level increase is accelerating due to intensifying climate change.

This cumulative perspective is particularly important because it demonstrates that even small annual increases can lead to substantial long-term consequences. Over a period of several decades, these incremental changes result in significant overall sea level rise.

The visualisation clearly shows that sea level rise is not a temporary or isolated issue but a growing and persistent global concern.

5.6. Regional ocean comparison

The dashboard also enables comparison of sea level trends across the Atlantic, Pacific, and Indian Oceans. The visualisations for each ocean show a consistent upward trend, indicating that sea level rise is a global phenomenon affecting all major ocean basins.

However, slight differences in the rate of increase can be observed between regions. These variations may be influenced by factors such as ocean currents, wind patterns, and regional climate conditions. For example, certain regions may experience stronger warming effects or changes in ocean circulation, leading to higher rates of sea level rise.

Despite these regional differences, the overall pattern remains consistent across all oceans. This suggests that while local factors may influence the magnitude of change, the underlying cause of sea level rise is global in nature.

The ability to compare regional trends provides valuable insight into how climate change impacts different parts of the world. It also highlights the importance of considering both global and regional perspectives when analysing environmental data.

5.7. Implications of findings

The findings from this analysis have several important implications. Firstly, the consistent upward trend in sea level rise confirms the ongoing impact of climate change on global ocean systems. This highlights the urgency of addressing climate-related issues to prevent further environmental damage.

Secondly, the presence of short-term variability demonstrates the complexity of climate systems. While fluctuations occur due to natural factors, they do not negate the long-term trend of increasing sea levels.

Thirdly, the cumulative nature of sea level rise emphasises the long-term consequences of climate change. Even small annual increases can lead to significant impacts over time, particularly for coastal regions. Rising sea levels increase the risk of flooding, erosion, and damage to infrastructure, which can have serious economic and social consequences.

Finally, the consistency of trends across different ocean regions suggests that sea level rise is a global issue that requires coordinated international efforts to address. Local solutions alone are not sufficient, as the underlying causes of sea level rise are global in scale.

5.8. Limitations

While the analysis provides meaningful insights into sea level trends, several limitations should be considered when interpreting the results. Firstly, the dataset is derived from satellite altimetry, which,

although highly accurate, is subject to measurement uncertainties such as instrument calibration errors, orbital drift, and atmospheric interference. These factors may introduce small variations in recorded sea level values.

In addition, the dataset represents global and regional averages, which may mask localised variations. Coastal regions, in particular, can experience significantly different rates of sea level change due to factors such as land subsidence, tectonic activity, and local ocean dynamics. As a result, the findings may not fully capture the complexity of sea level behaviour at smaller spatial scales.

Another limitation lies in the reliance on historical observational data without incorporating predictive modelling. While trend lines and moving averages provide useful approximations of long-term behaviour, they assume relatively stable patterns and do not account for potential nonlinear acceleration driven by climate feedback mechanisms.

Furthermore, the analysis focuses primarily on sea level measurements and does not integrate other key climate variables such as temperature anomalies, greenhouse gas concentrations, or ice sheet mass balance. The exclusion of these variables limits the ability to establish stronger causal relationships between observed trends and underlying climate drivers.

Therefore, while the visualisations effectively demonstrate observable trends, the results should be interpreted as indicative rather than definitive, and further analysis incorporating multi-variable climate models would provide a more comprehensive understanding.

5.9. Summary

In summary, the results clearly demonstrate a consistent and accelerating trend in global sea level rise. The combination of multiple visualisation techniques provides a comprehensive understanding of both long-term trends and short-term variability.

The findings highlight the importance of continued monitoring, data analysis, and climate action to address the growing challenges associated with rising sea levels.

Conclusion

This project successfully developed an interactive data visualisation dashboard that explores global and regional sea level changes between 1993 and 2024. The dashboard combines multiple visualisation techniques to provide insights into long-term sea level trends, yearly variations, and regional differences.

The results clearly demonstrate that global sea levels have been rising steadily over the past three decades. This trend is consistent with scientific research that links sea level rise to climate change, ocean warming, and melting ice sheets.

Interactive visualisation tools such as Streamlit and Plotly provide an effective way to explore climate datasets and communicate complex environmental information. By allowing users to interact with the data, the dashboard helps improve understanding of climate change trends.

Future improvements to this project could include integrating additional climate datasets such as temperature anomalies, glacier melt rates, or carbon emissions. Incorporating predictive models could also help estimate future sea level rise and provide additional insights into the long-term impacts of climate change.

References

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Appendix

Appendix A (Charts)



Fig 3
(Global Sea Level Rise Chart)

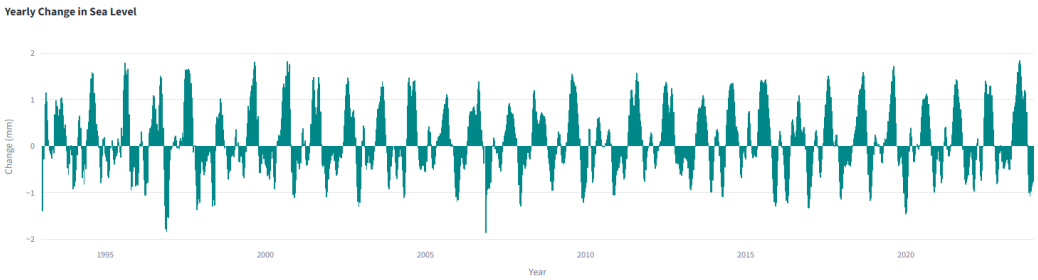


Fig 4
(Yearly Change in Sea Level Chart)

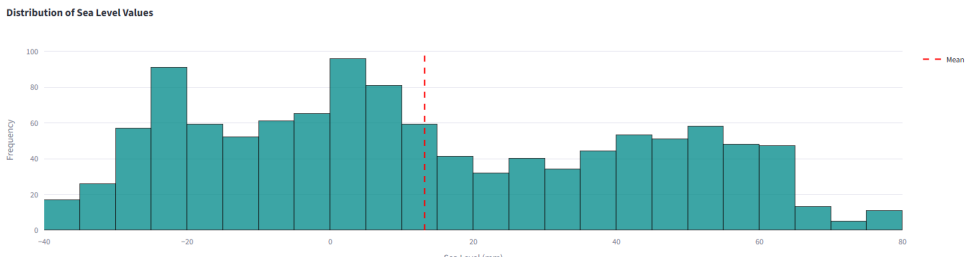


Fig 5
(Distribution of sea level values chart)

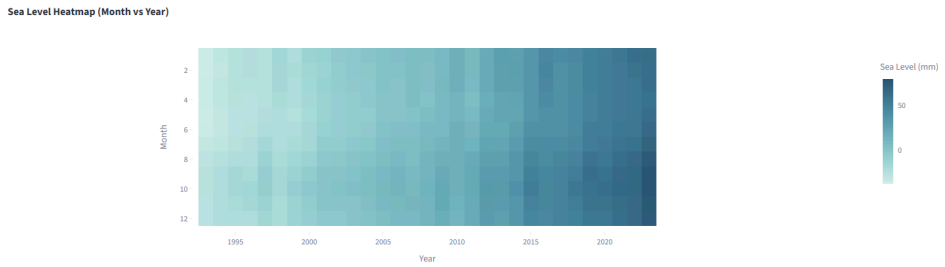


Fig 6
(Sea Level Heatmap Chart)

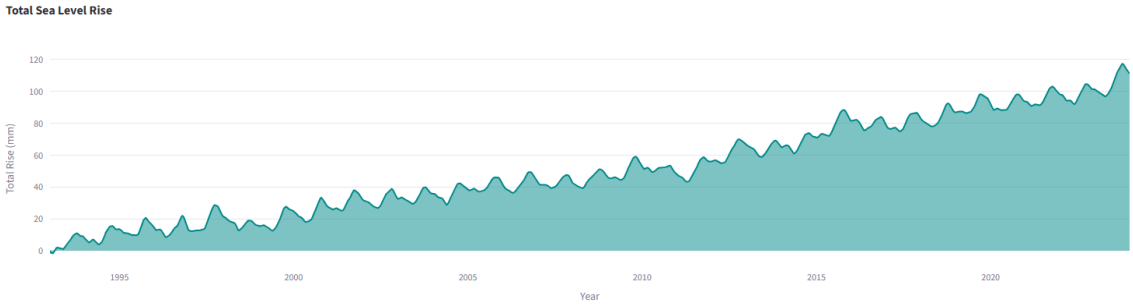


Fig 7
(Global Total Sea Level Rise)

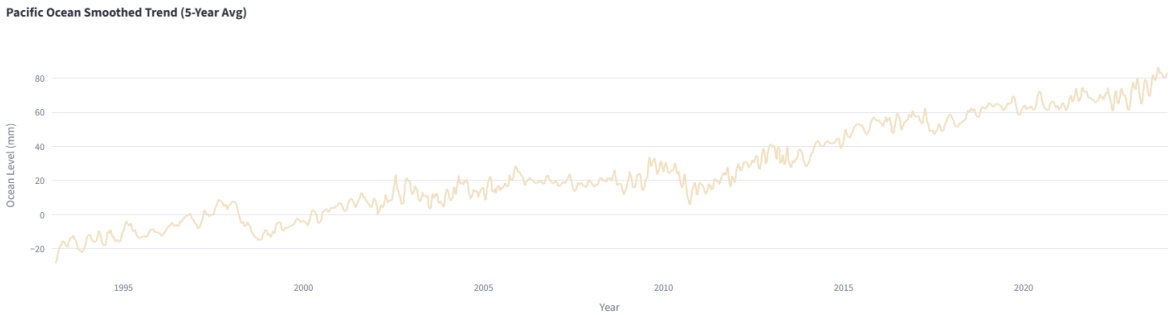


Fig 8
(A line chart of a smoothed trend for ocean basins)

Appendix B (Code Implementation)

```
df["YearlyChange"] = df["SmoothedGMSLWithGIA"].diff()
df["Acceleration"] = df["YearlyChange"].diff()
df["CumulativeRise"] = df["SmoothedGMSLWithGIA"] - df["SmoothedGMSLWithGIA"].iloc[0]

df["Year"] = df["YearPlusFraction"].astype(int)
df["Month"] = ((df["YearPlusFraction"] % 1) * 12 + 1).astype(int)

coeffs = np.polyfit(df["YearPlusFraction"], df["SmoothedGMSLWithGIA"], 1)
trend = np.poly1d(coeffs)
df["Trend"] = trend(df["YearPlusFraction"])
df["RollingAvg"] = df["SmoothedGMSLWithGIA"].rolling(window=12).mean()
```

Fig 9

(Feature engineering process used to compute yearly change, cumulative rise, trend line, and moving average for sea level analysis)

```
fig1 = px.line(
    filtered,
    x="YearPlusFraction",
    y="SmoothedGMSLWithGIA",
    title="Global Sea Level Rise",
    labels={
        "YearPlusFraction": "Year",
        "SmoothedGMSLWithGIA": "Sea Level (mm)"
    },
    color_discrete_sequence=["darkcyan"],
    hover_data={
        "Year":True,
        "Month":True,
        "SmoothedGMSLWithGIA":":.2f"
    }
)
```

Fig 10

(Implementation of the global sea level line chart using Plotly Express to visualise time-series data)

```
fig1.add_scatter(
    x=filtered["YearPlusFraction"],
    y=filtered["Trend"],
    mode="lines",
    line=dict(color="red", dash="dash"),
    name="Trend",
    hovertemplate="Year: %{x:.0f}<br>Month: %{customdata[1]}<br>Trend: %{y:.2f} mm",
    customdata=filtered[["Year","Month"]]
)

fig1.add_scatter(
    x=filtered["YearPlusFraction"],
    y=filtered["RollingAvg"],
    mode="lines",
    line=dict(color="orange"),
    name="Moving Average",
    hovertemplate="Year: %{x:.0f}<br>Month: %{customdata[1]}<br>Moving Avg: %{y:.2f} mm",
    customdata=filtered[["Year","Month"]]
)
```

Fig 11

(Addition of regression trend line and moving average to enhance interpretation of long-term sea level trends)

```
dataset = st.sidebar.selectbox(
    "Select Ocean",
    ["Global", "Atlantic", "Pacific", "Indian"],
    format_func=lambda x: {
        "Global": "Global",
        "Atlantic": "Atlantic Ocean",
        "Pacific": "Pacific Ocean",
        "Indian": "Indian Ocean"
    }[x]
)

st.sidebar.markdown("""
You can analyse:

• **Global sea level rise**
• **Atlantic Ocean trends**
• **Pacific Ocean trends**
• **Indian Ocean trends**
""")

st.sidebar.markdown("""
Adjust the year range to observe how sea levels change over time.
""")

year_range = st.sidebar.slider(
    "Select Year Range",
    1993,
    2024,
    (1993, 2024)
)
```

Fig 12

(Implementation of interactive dashboard controls allowing users to filter data by ocean region and year range)